

Fluid Catalytic Cracking

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Javier Eduardo Mendieta
Omar Zahdan
Christian Sanchez
Fatimah Al Ibrahim

Table of Contents:

	Page#
Project Charter	2
Design Plan	3
Introduction	4-5
Market Analysis	
- Feedstock	6-9
- Final product	9
- Competitors and Companies	10
- Effect of sulfur regulation on the Health benefits	10
Other Possible Processes	
- Isomerization	11-12
- Steam Cracking	12-13
- Thermal Cracking	13
- Hydroprocessing	14 -15
The Chosen Solution	
- Chosen Solution	15-16
- What the process is	16-27
- Chosen sulfur treatment	28-30
Appendix I: Design Basis	31
Appendix II: BFD and material balance	31-32
Appendix III: PFD	33-35
References	36-39

Project Charter:

TITLE	GOAL	IN SCOPE	OUT OF SCOPE
Fluidized Catalytic Cracking	To Produce High Octane Gasoline with Low Levels of Sulfur in the FCC process.	- BFD, PFD, and Material Balance - Judging Criteria - Possible Solutions - Design Basis	- Process Control - Equipment Sizing - Equipment Cost - Final PFD - Heat Integration
DELIVERABLES		TEAM MEMBERS	
- Market Analysis - Process Description - Solutions to the Problem - Design Basis		Javier Eduardo Mendieta Omar Zahdan Christian Sanchez Fatimah Al Ibrahim	

Design Plan:

Homework Problems Design Plan					
Task ID	Task	Start Date	Duration (days)	Finish Date	Assignee
Logistics					
1	Task Delegation	24-Jan	1	24-Jan	Everyone
2	Project Charter	24-Jan	1	24-Jan	Javier Eduardo
3	Design Plan	24-Jan	4	24-Jan	Javier Eduardo
4	Layout/Format	24-Jan	9	1-Feb	Everyone
Execution					
5	Definition of Big Problem	24-Jan	5	28-Jan	Fatimah
6	Why is this problem important?	24-Jan	5	28-Jan	Fatimah
7	Explanation of Goals and Objectives	24-Jan	5	28-Jan	Fatimah
8	Market Analysis	24-Jan	5	28-Jan	Christian
9	Judging Criteria	24-Jan	5	28-Jan	Christian
10	What is the process?	24-Jan	5	28-Jan	Javier Eduardo
11	What are the possible solutions?	24-Jan	5	28-Jan	Javier & Omar
12	What is the solution?	24-Jan	5	28-Jan	Javier Eduardo
13	Why have we chosen this solution?	24-Jan	5	28-Jan	Omar
13	The chosen sulfur treatment	24-Jan	5	28-Jan	Fatimah
14	Design Basis	24-Jan	5	28-Jan	Omar
15	BFD & Material Balance	24-Jan	5	28-Jan	Everyone
16	PFD	24-Jan	5	28-Jan	Fatimah
Wrap-up					
17	Section Unification	1-Feb	1	1-Feb	Everyone
18	Proofreading	1-Feb	1	1-Feb	Everyone
19	Reference Page	1-Feb	1	1-Feb	Everyone

Introduction:

In the present days, gasoline, which is a mixture of hundreds of light hydrocarbons ($C_4 - C_{12}$), is considered as one of the most consumed fuel in the United States. According to the U.S. Energy Information Administration, Americans used around 400 million gallons of gasoline per day in 2018[1]. This represents 47% of the total finished petroleum-based fuel consumption in the U.S. as figure 1 demonstrates [2]. Almost 92% of gasoline is consumed as engines for light-duty vehicles such as cars and small trucks [1]. Due to the high demand of gasoline in the United States, the purpose of this project is to produce high octane gasoline with low sulfur concentration.

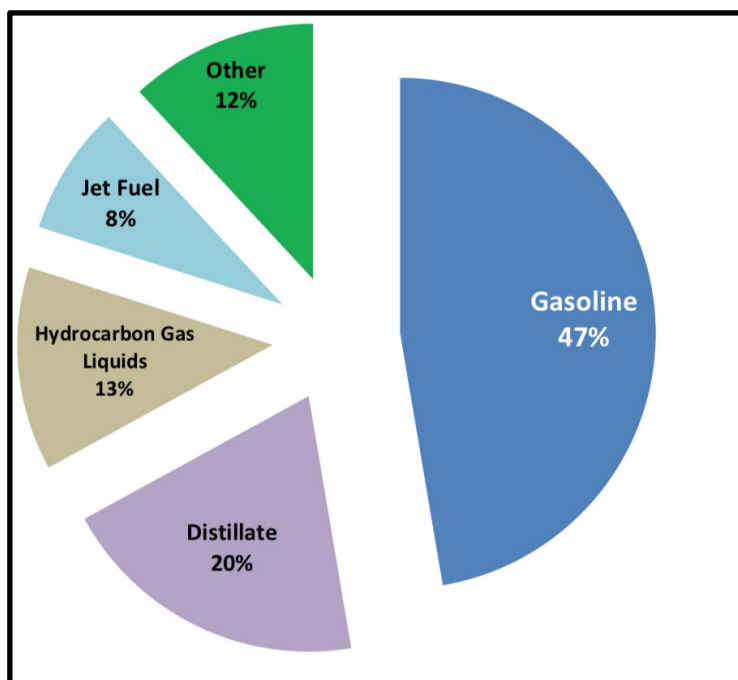


Figure 1: The U.S. Petroleum-Based Consumption in 2016 [2]

Octane rating plays an important role in classifying the gasoline grade since it considers of the main property of gasoline. There are three classifications of gasoline regarded as 87

Regular, 89 Plus, and 91-93 Premium [3]. Octane rating measures the amount of fuel that can be compressed before it spontaneously ignites. In other words, it measures the quality or the performance of gasoline as fuel. The higher the octane number is, the better the performance of the fuel is. For instance, the resistance of gasoline mixture to knock is higher when the octane number is high. In fact, engine knocking is known as pre-ignition. When the air-fuel mixture inside the cylinder ignites because of the compression instead of the spark plug, engine knocking occurs [4]. There is a proportional relation between the compression and the octane number. Thus, the higher the octane number, the more compression can be resist before ignite and then less engine knocking can occur.

Despite the importance and uses of gasoline, the combustion of gasoline is considered as one of the main sources of air pollution due to the production of emissions of several harmful substances such as CO, NO_x. Therefore, many countries in the world such as South Korea, Europe and the U.S. have new regulations for sulfur in gasoline. In the United States, on an annual average basis, the regulation for sulfur in gasoline is no more than 10 parts per million (ppm) [2]. Furthermore, the new IMO (International Maritime Organization) limits on sulfur is 0.50% m/m (mass by mass) which has been applied since January 1st, 2020 [5]. Although the new regulation will help reduce the emission of sulfur gasoline, octane gasoline will be lower. Therefore, the purpose of this project is to produce low sulfur gasoline with octane retention.

Market Analysis:

- Feedstock

Since the new regulations on sulfur emissions of gasoline have begun on January 2020 there is a mandatory change that needs to be implemented on to the FCC unit. The FCC unit will then need to either have purified crude oil or use an oil with less sulfur content like Tight Oil to meet the regulation of lower sulfur emissions. Tight oil also referred to as shale oil, or LTO could substitute the heavy crude oil for the production of gasoline. Focussing only on the United States, oil production is 11 million barrels a day in 2018 [40]. To compare, the top 5 oil producing countries of more than half of the production of oil can be seen in figure 2 below.

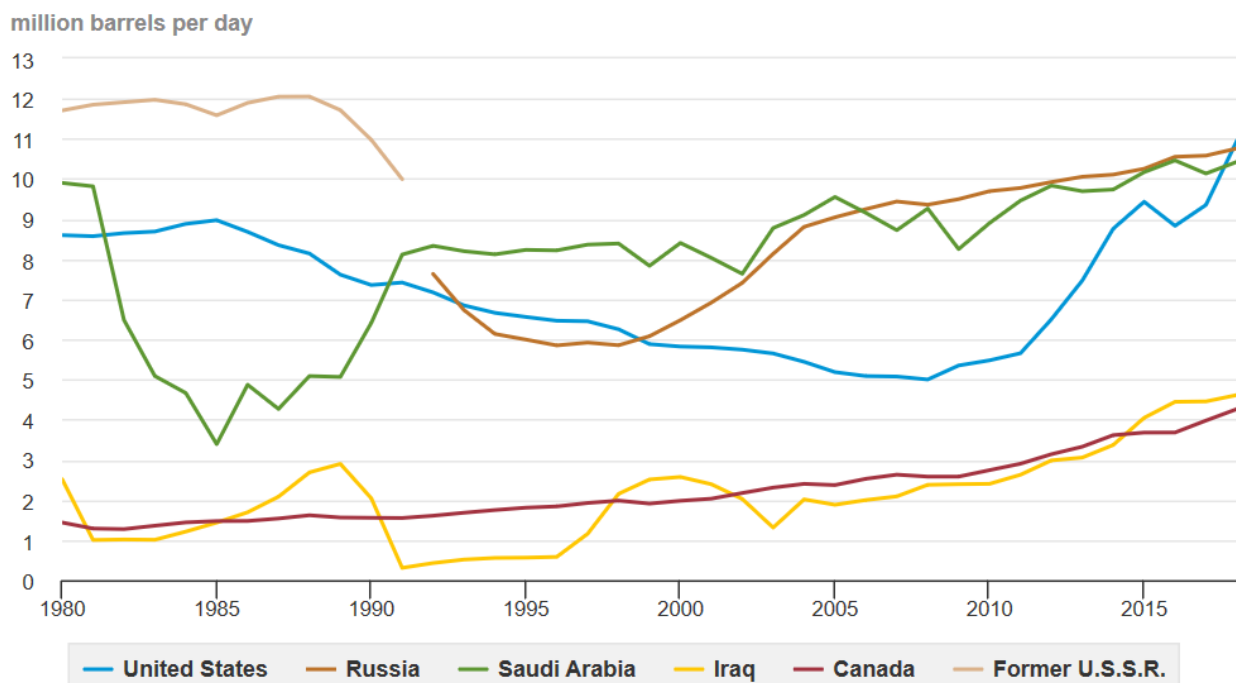


Figure 2 - Top five crude oil Producing countries, 1980-2018 [40]

As seen in figure 2, the production of oil is increasing, thus more gasoline can be produced, yet the new restrictions on sulfur emissions makes the solution to this problem relevant. Since the

United States has the highest potential of production of oil in the near future the market analysis will be focused on this country.

Crude oil is a necessity when producing gasoline, though there are different grades of crude oil with a difference in sulfur content in the crude mixture. The largest amount of crude oil produced in the United States being LTO, as stated before, this feedstock has the lowest amount of sulfur in the crude mixture making it an appropriate candidate to be used in an FCC unit. The variance in the production of oil in the U.S. can be seen in figure 3 below.

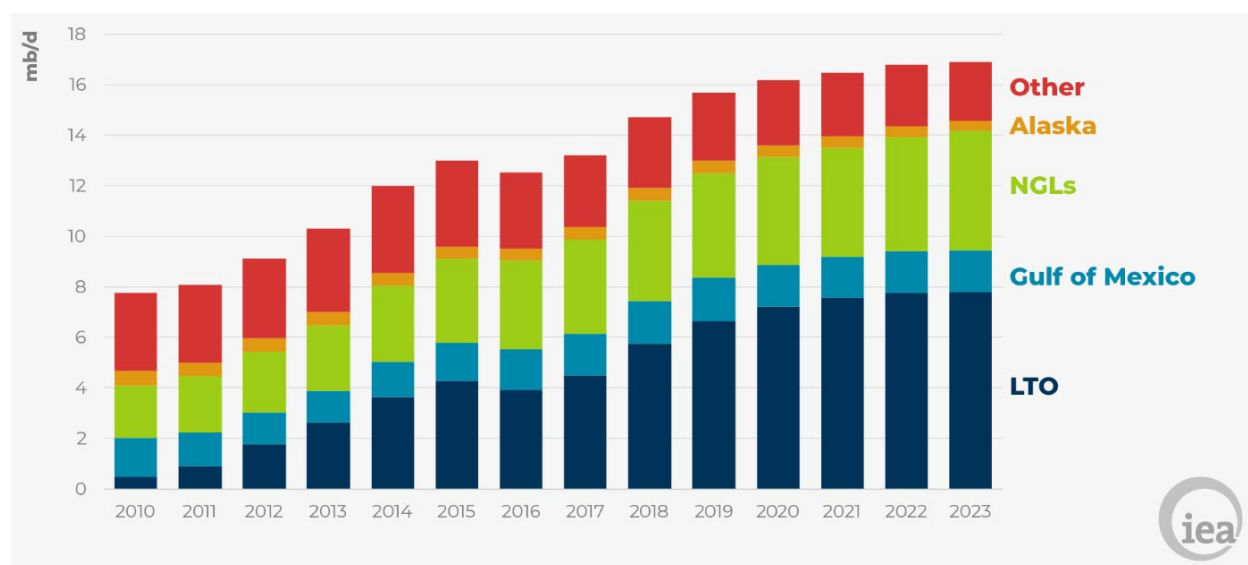


Figure 3 - Forecast of United States Oil supply from 2010-2023 [41]

Analyzing figure 3, the projection of LTO is steadily rising as the years progress; thus being a useful resource to the FCC unit to produce cleaner gasoline.

Exportation of the gasoline from the U.S. can show why the importance of cleaner production of gasoline to other locations around the world must be met. Since there is an increasing demand of gasoline around the world, a solution to the IMO regulation must be found, observing the exports of gasoline from 2015 to 2019 had a steady oscillating increase of exports,

as low as 400 million BBL of gasoline a day in 2019 to 1000 million BBL/day [42]. The described trend can be seen below in figure 4.

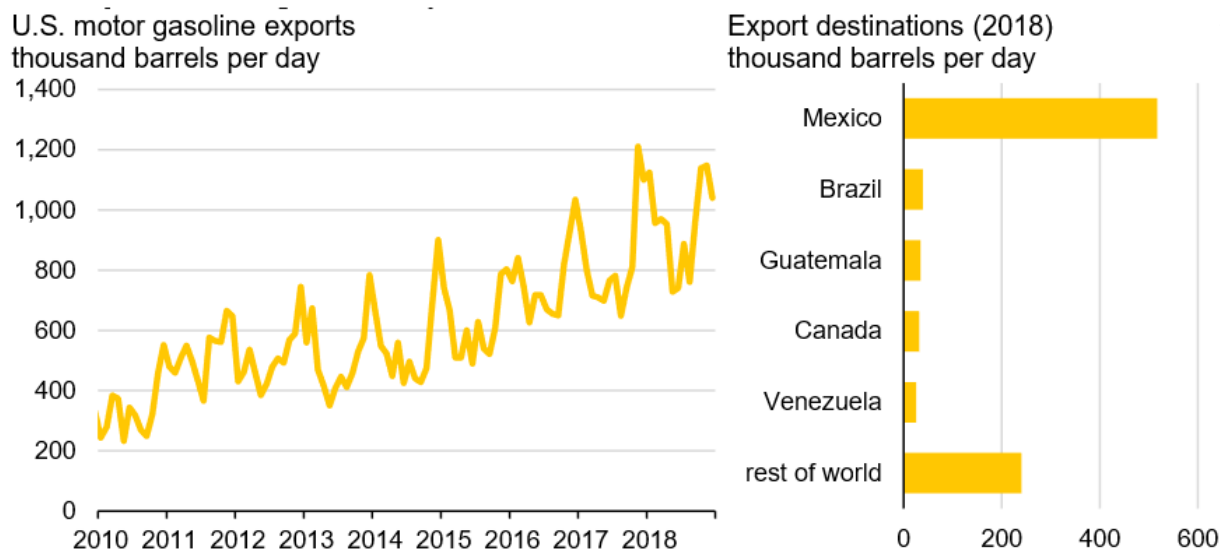


Figure 4 - United States exports of gasoline from 2010 to 2019 [42]

In a typical FCC unit, the feedstock used is vacuum gas oil (VGO) mixed with coker off gas (COG). The VGO market growth is 5.2% in the United States, with a market size of 338 billion USD in 2018. Due to the high demand of gasoline and diesel, the market size is expected to reach 510 billion USD by 2026. On the other hand, if VGO is pretreated before entering the FCC unit, the market growth is lower at 0.5 to 0.7%, with a market size of 220 billion USD in 2018. Thus, VGO that is not treated is higher in demand. This is because most refineries prefer to treat the feed themselves to make it more economically feasible and profitable [44]. As of January 2020, the cost of low sulfur VGO is \$14.75/BBL. Compared to the price of high sulphur VGO at \$13.5/BBL [45]. Having a higher cost VGO as a feed to the reactor means that the prices of consumer gasoline will also rise.

The coker off gas is neither bought or sold in the market. Rather it is a product that is produced from the delayed coker which is refeed into the FCC unit to extract any residual hydrocarbons [44].

- Final Product

The final product of this process is low sulfur gasoline. In the United States, the average price of gasoline is \$2.50 per gallon. The highest price of gasoline across the country is in San Francisco, California with net value of \$3.45 per gallon, whereas the lowest price of gasoline is in St. Louis, Missouri at a price of \$2.10 per gallon.

The trend for the prices of gasoline from 2008 to 2018 can be seen in figure 5 below.

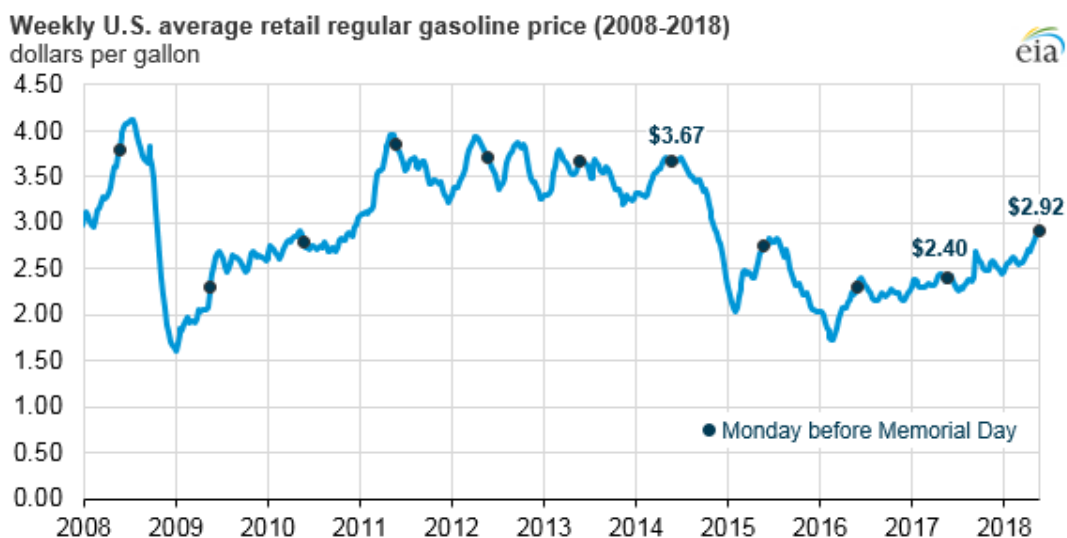


Figure 5 - Weekly U.S. average retail regular gasoline prices from 2008 to 2018 [45]

- Competitors and Companies

In the United States, the biggest refineries and competitors to our project are Marathon Petroleum, Valero Energy, Exxon Mobil, Phillips 66, and Chevron. Marathon Petroleum currently has 16 refineries across the country, with a production of three million barrels per day. Valero Energy has 13 refineries with a production of 2.2 million barrels per day. Exxon Mobil currently has five refineries in the United States but it manages to produce 1.7 million barrels per day. Phillips 66, on the other hand, has nine refineries but a production of 1.5 million barrels per day. For example, Chevron owns five refineries in the United States with a production of 1.1 million barrels per day [46].

- Effect of sulfur regulation on the Health benefits

Although lowering sulfur content in gasoline will higher the price of gasoline due to the technology required to lower sulfur content in gasoline, there are many benefits behind low sulfur gasoline on the overall market. For instance, reducing sulfur gasoline will help significantly reduce gasoline emission, therefore, reducing the air pollution. In fact, due to air pollution, there are approximately 23,000 cases of children's respiratory illnesses. Furthermore, the premature deaths cases are estimated to be around 2400 annually [39]. As a result, lowering the sulfur content in gasoline will significantly reduce the fuel vapor' emission to near zero; thus, the health benefits from the new sulfur regulation is estimated to be annually around \$8 - \$23 billion by the year 2030 [39].

Process Description:

- **Competing Processes**

- **Isomerization:**

Isomerization is the chemical process in which one molecule is chemically rearranged into another molecule with the exact number of atoms (isomers). In petrochemical refineries isomerization is used to convert straight chained alkanes into branched chained alkanes with the purpose of improving the octane rating of gasoline [7]. Due to the chemical composition of gasoline, the targeted alkanes are butane, pentane, and hexane. The feedstock for isomerization comes from light naphtha [8]. These compounds are treated in the presence of a catalyst which alternates depending on the operating conditions.

If isomerization takes place under low temperature conditions, aluminum chloride (AlCl_3) is used in the presence of hydrogen chloride (HCl) to isomerize primarily butane and prevent the formation of olefins or alkenes. The temperature range for this process goes from 230 to 340°F while the operating pressure goes from 200 to 300 psi [7]. On the other hand, the operating temperature is higher when isomerization of pentane and hexane is targeted. In this case, metal catalysts such as platinum are used. Given that pentane and hexane increase the octane rating of gasoline, these compounds are usually isomerized in petrochemical industries [9]. One of the advantages of isomerization is that the products are already desulfurized and dried, which makes of isomerization a highly efficient process for gasoline production. Nevertheless, if the feedstock was not dried and properly washed with water and caustic, there are high possibilities of acid formation [7]. Thus, leading to corrosion of units and catalyst poisoning. Additionally, the usage of HCl as catalyst requires the appropriate piping system for draining [8].

The following figure shows how the rearrangement of atoms can take place for pentane.

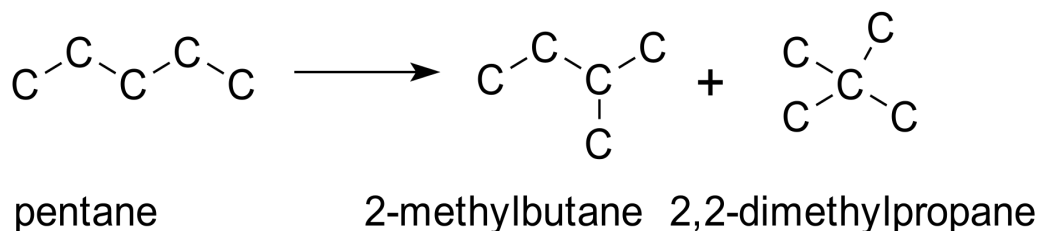


Figure 6 - Isomerization of pentane [11]

- **Steam Cracking:**

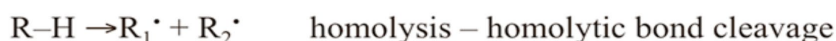
Steam cracking is a petrochemical process that breaks down alkanes into smaller hydrocarbons by diluting the feedstock with steam and allowing it to enter the furnace at very high temperatures (around 800°C) [32]. Recently, the residence time allowed for the reaction to take place in the furnace is very small, only amounting to a couple milliseconds but still reaching the cracking temperature. The low residence time is important to ensure a high yield of desired product. After cracking, the stream is sent to a quench tower to reduce the temperature rapidly and avoid formation of side products [35]. Depending on the feedstock being used in the cracker this process is capable of generating many products including aromatics, light alkenes, and alkanes. If a light feedstock is used such as natural gas or light naphtha, the process will produce light alkenes such as propene [30]. However, when heavy naphtha is used as a feedstock, the steam cracking process is capable of generating alkanes that can be added to gasoline making steam cracking a valid alternative for gasoline production [29]. Since steam cracking is a form of thermal cracking it follows the same mechanism.

- **Thermal Cracking:**

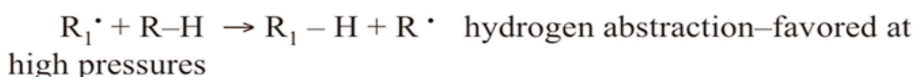
Thermal cracking is a petrochemical process that breaks down long chain hydrocarbons into more useful smaller chains [21]. This process occurs at a high temperature range between

750 °C and 900 °C and a high pressure of around 7000 kPa. It is also the oldest form of cracking, steam cracking is a form of thermal cracking [19]. However, in thermal cracking the feed is not diluted with steam. The addition of steam can reduce coke formation and increase the yearly runtime of the cracking unit [22]. This process also does not use a catalyst which means that high temperatures are required for the cracking process to take place just like the steam cracking process. Figure 7 below displays the chemistry of the thermal cracking reaction which occurs via a radical mechanism.

Initiation



Propagation



Termination

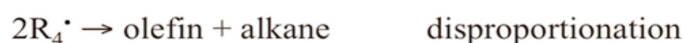


Figure 7: Radical Mechanism of Thermal Cracking Reaction [20]

- Hydroprocessing:

Hydroprocessing describes both hydrocracking and hydrotreating which are similar in some aspects and different in other aspects. Hydrocracking is a form of catalytic cracking that uses the light product of the distillation, called VGO as a feed. This feed is obtained in the vacuum unit following the distillation of crude oil [21]. It uses the same feed as the FCC unit, however, this feed is injected with hydrogen to obtain a higher quality diesel and kerosene. VGO is not the only feed to the hydrocracker but it is the most commonly used in industry [28]. This process also produces gasoline but the main objective when using this process over FCC, is the

production of good quality diesel and kerosene [22]. Gasoline is a product that is produced in lesser quantities. The main purpose of hydrocracking is to break long chain hydrocarbons into more useful short chain hydrocarbons, just like the process of steam and thermal cracking. In this process, the presence of catalyst is required and it's referred to as the hydrotreating catalyst [19]. This catalyst increases the rates of the reactions involving sulphur and nitrogen removal. Also, the catalyst increases the saturation rate of the hydrocarbons [22]. This removal of sulfur and nitrogen occurs in the hydrotreater before being collected as a final product or used at another unit. Hydrotreaters operate at temperatures between 500°F and 750°F and mixed with hydrogen just like in the hydrocracker [23]. Usually, the catalyst used in this process is nickel molybdenum or cobalt molybdenum [28]. With the addition of this catalyst, the sulfur becomes hydrogen sulfide gas while the nitrogen is turned into ammonia [26]. Depending on the feed, the operating pressure of the hydrotreater is adjusted. The heavier the feedstock the higher the operating pressure of the unit [23].

An overall flow diagram of the hydrocracking process is shown in figure 8.

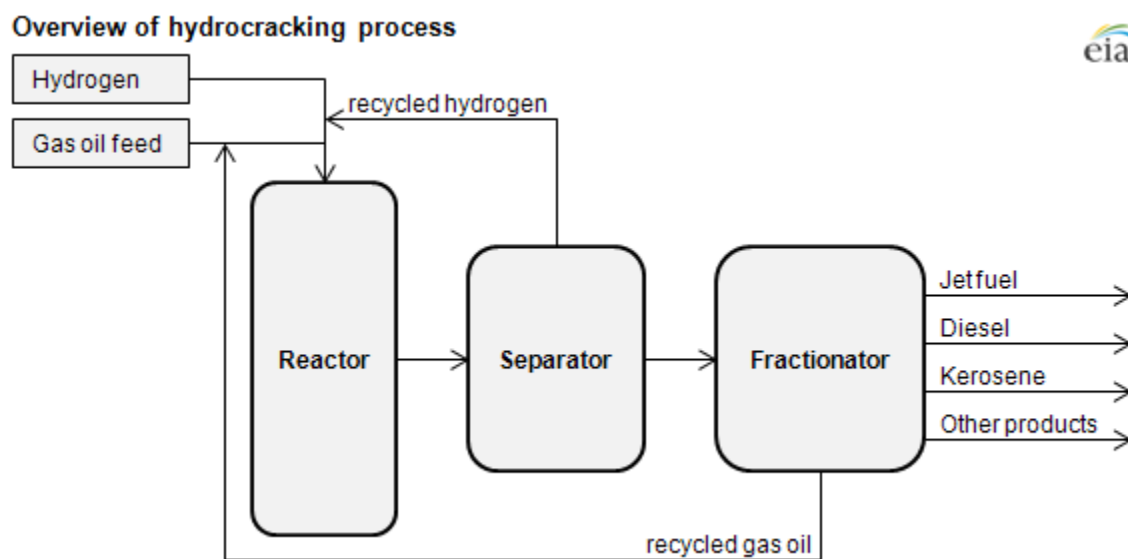


Figure 8 - Overall Diagram for the Hydrocracking Process [26]

Chosen Solution:

The production of high octane and low sulfur gasoline can be achieved by many processes such as isomerization, cracking, and reforming [29]. There are several types of cracking such as catalytic cracking, thermal cracking, steam cracking, and hydrocracking. After analyzing the pros and cons of every process it was decided that the most optimal process for producing gasoline would be through Fluidized Catalytic cracking (FCC)[28]. When FCC is compared with thermal and steam cracking, we observe the advantage of using FCC over thermal cracking and steam cracking. The use of a catalyst means that lower temperature and pressure is required to reach cracking conditions, allowing the plant to minimize energy consumption costs and increase profitability[31]. It also means that the rate of desired and undesired reactions can be altered by changing the type of catalyst thus improving the selectivity of the final product. Furthermore, the thermal efficiency of the FCC is greater than thermal cracking [34]. This is due to the thermal recovery of heat from the combustion reactions that takes place in the regenerator to remove the coke formation from the FCC catalyst [33]. The FCC process also produces gasoline with a high octane rating because the process produces, low normal alkane yields and high yields of isoalkanes and aromatics which increase the octane rating [29]. On the other hand, thermal cracking produces low octane gasoline and low chain branching in alkanes. This makes FCC a more viable option when producing gasoline as a high octane rating is required to be able to resist knock and be a desired product to the consumers [32].

Although isomerization is an efficient process for ensuring a high octane rating gasoline and low sulfur levels to meet present day regulations. The main drawback for using this process includes the low compositions of water, sulphur and aromatics in the feed that is required for the

isomerization process [35]. Therefore, VGO is not suitable for the isomerization process because it contains high levels of sulphur and aromatics and cannot be used as a feedstock unless we resort to cracking. Therefore, FCC is the optimal choice when we are given this type of feed [29].

- What is the process?

In most refineries, the main goal is to process and break down the crude oil to refine it into petroleum derived products that are more applicable to suit the necessities of today's world. These products include: naphtha, gasoline, kerosene, heating oil, diesel, asphalt, and liquefied petroleum gas (LPG) [10]. To achieve this goal, refineries have been designed with many processing units that will convert the crude oil into these products [12].

In a typical refinery, the first unit used for this process is the crude unit, which is a distillation column that operates under atmospheric conditions. In the crude unit, the stream of raw crude oil enters the tower to be distilled into different raw products such as raw kerosene, raw diesel, and gas oil. The bottoms of the tower cannot be further separated under atmospheric conditions; thus, the heaviest portion of the crude oil is heated up to a temperature of 400 degrees Celsius (in a fired furnace) and sent down to the vacuum unit [14].

The vacuum unit is a set of distillation columns that operate under pressures below 1 atm. Under these conditions, the boiling point of the hydrocarbons lowers until they can be separated without degrading the oil as it would happen if they were treated at higher temperatures [15].

From the vacuum distillation three main sets of products are obtained; the light gas oil, heavy gas oil, and the tar or residues. Normally, the light and heavy gas oil streams are sent to the FCC unit, while the residues are sent to the delayed coker [17]. The delayed coker is considered the process unit with the most extreme conditions since it cracks the tar into lighter

hydrocarbons and solid carbon. The lighter hydrocarbons are sold as fuel gas, whereas the solid carbon can be further processed into the deasphalting unit, where asphalt is obtained and sold for the construction of roads [16]. The overall layout of this section of the process is displayed in the following figure.

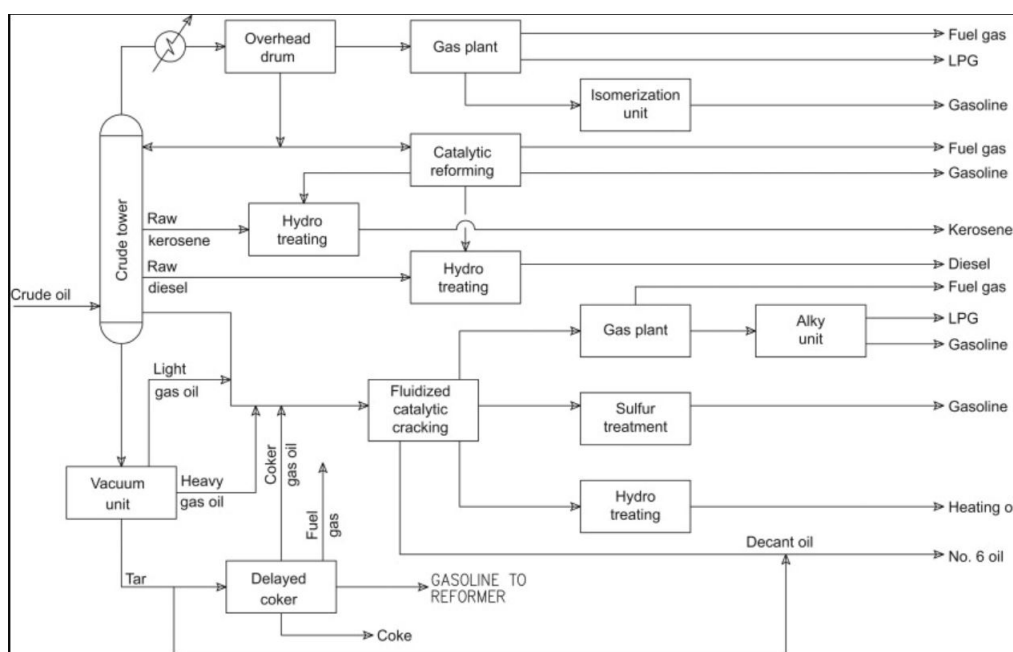


Figure 9 - Refining Overall Process [16]

As shown in figure 8, the lighter products from the bottoms of the crude tower are sent directly to the FCC unit. However, the feed stream is preheated and then sent through the feed nozzles or riser, catalyst separation, stripping section, regenerator, main fractionator, the gas plant, and the treating facilities, which make up the FCC unit [15].

During the pre heating section of the FCC process, the feed preheat temperatures range from 205°C to 400°C. To reach these temperatures, the stream goes through a series of heat exchangers that use the hot streams from the fractionator of the FCC unit for heating purposes. After the gas oil stream has passed through the heat exchangers, the stream goes to a specialized furnace that will maximize the feed temperature [18].

The next section of the FCC unit is the riser. This is the heart of the FCC process because it consists of the reactor. The reactor is equipped with nozzles that maximize the contact of the feed with the catalyst, enhancing the desired cracking reactions. Additionally, the nozzles are used to atomize steam incrementing the number of reactive sites of the catalyst to promote the cracking reactions [14].

In the reactor, the catalytic reactions initiate as soon as the feed is vaporized by the hot regenerated catalyst [1]. The driving force used for transportation of the catalyst to the reactor is the expanding volume of the gases (as temperature rises, the volume of gases increases). The temperature of the catalyst ranges between (670°C to 730°C), giving an overall reactor temperature of 520°C [12]. Due to this temperature range, the cracking reactions take place because catalytic cracking is an endothermic process. Thus, the rise in temperature favors it [15].

In the FCC unit, the riser consists of a vertical pipe (typically 75 to 125 ft long and 3 to 8 ft in diameter), which resembles a plug flow reactor. This resemblance comes from the fact that back mixing is minimized while the catalyst and the vapors travel through the pipe at a speed of 50 ft/s, approximately [17]. Due to these features, the residence time between the catalyst and the hydrocarbons is about 2 to 3 seconds, which unfortunately leads to the formation of solid carbon on the surface of the catalyst. Thus, the catalyst activity is reduced [18].

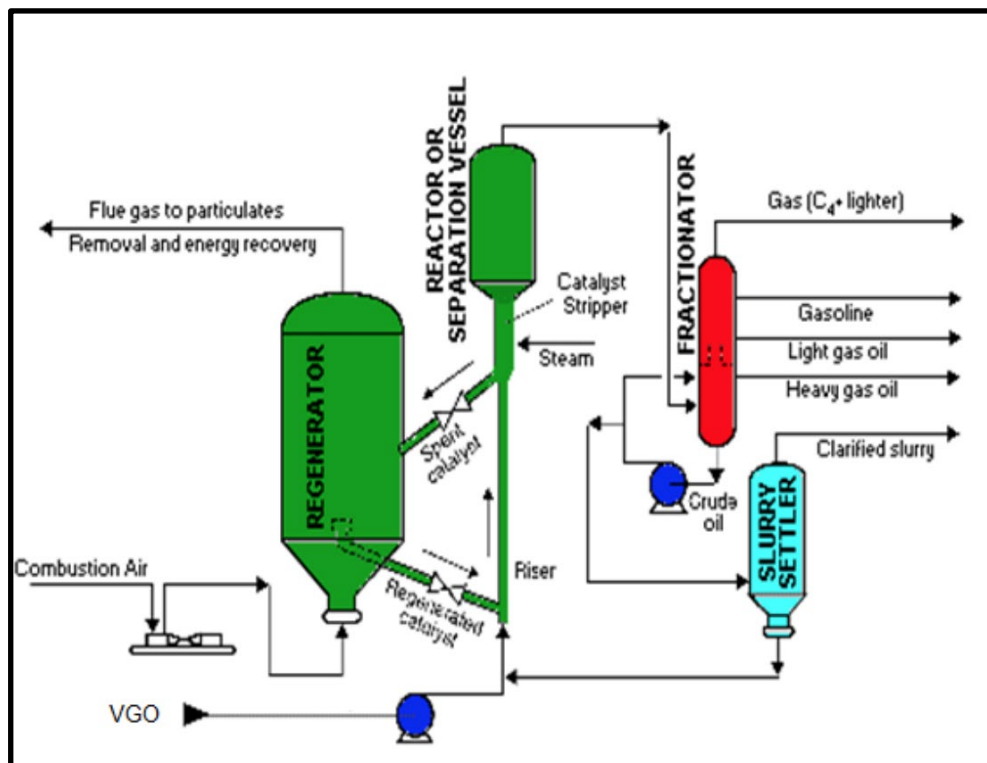


Figure 10 : The process flow diagram of FCC process [12]

After the catalyst exits the riser, it enters the separation or reactor vessel as seen in figure 10. In this reactor vessel, the bulk catalyst is separated from the vapors in a complex system known as cyclones. As soon as the mixture of catalyst and vapors enter the cyclone, it encounters a deflector device which sends the catalyst downwards and the vapors upwards. Due to this mechanism, an average separation of 88% can be achieved depending on the driving force of the cyclone scheme [14]. Therefore, many refineries have opted to implement systems of at least two stages-cyclones, achieving a nearly perfect separation of 99.9%. After separation is completed, the vapors are sent to the main fractionator tower, while the catalyst is sent to the stripper through the trickle valves or dipleg. Separation in the cyclone is essential for the FCC process because longer contact with the catalyst may lead to the formation of undesired products [13]. Figure 11 below displays an accurate representation of how the cyclone tower operates.

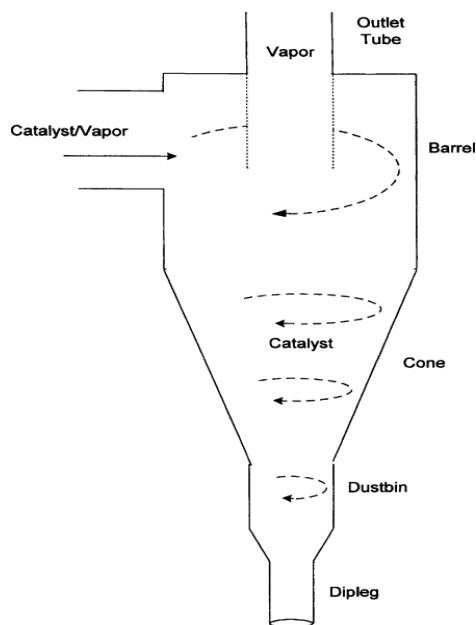


Figure 11 - Catalyst/Vapor Separation in a Cyclone System [12]

In the stripping section, the catalyst coming from the cyclone is exposed to stripping steam. This steam is used to remove any hydrocarbons that may have been brought along with the catalyst particles [14]. However, due to the structure of the catalyst particles, some hydrocarbons may be stuck to the catalyst's pores, which cannot be removed by applying the stripping steam onto the catalyst. Any stripping section for FCC is required to have a steam velocity of 0.75 ft/s, providing an optimal contact between the catalyst and the steam. If the contact were to be longer than required, the catalyst would start to entrain with the steam, thus no efficient separation would occur [18].

Since the stripping section is at a temperature of 510°C, cracking reactions still take place converting the long chained hydrocarbons into lighter vaporized chains that will be sent up through the cyclone to the fractionator [17]. Figure 12 shows the stripping section of the FCC unit and the flow of steam from the bottom of the stripping section to the top.

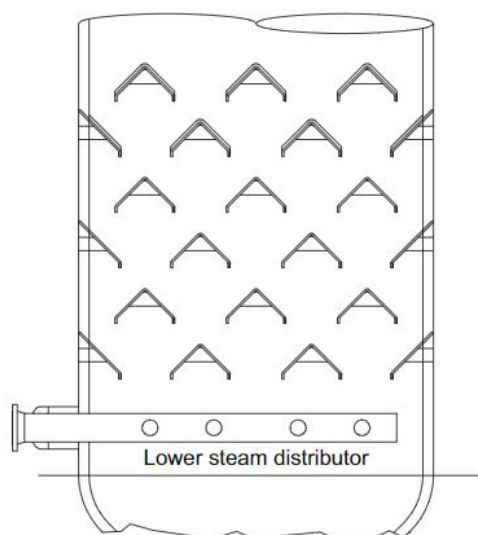


Figure 12 - Stripping Section in the FCC Unit [12]

After the catalyst has been efficiently separated from the entrained hydrocarbons, the catalyst is sent to the regenerator. At this point of the process, the catalyst is known as ‘spent catalyst’ because it still has some deposits on its surface [17]. These deposits cover 1.5% of the overall catalyst particle and it mainly consists of coke. Coke is chemically defined as a layer of solid carbon also containing some hydrogen, a small traces of other organic compounds such as amines and amides, and sulfur [16].

In the regenerator, coke is removed from the catalyst through a series of combustion reactions that use air as a source of oxygen. The flow of air enters the regenerator through air blowers whose functions are to keep the catalyst in a fluidized state by providing the adequate pressure and inflow of air. Consequently, the main goals of the regenerator are to restore the catalyst activity, to supply heat for the cracking reactions (combustion is exothermic), and to deliver the fluidized catalyst to the feed nozzles at the entrance of the riser [14].

Due to the chemical composition of coke, the following reactions take place when air is let into the regenerator:

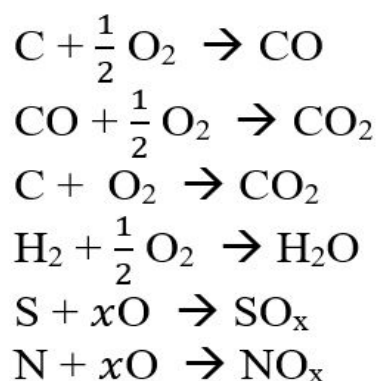


Figure 13 - Combustion Reactions required to remove Coke [13]

These combustion reactions can take place partially or completely depending on the limiting reagent. When partial combustion takes place, the limiting reagent is oxygen, thus not all the coke can be removed from the catalyst and less energy is generated to supply the cracking reactions [17]. On the other hand, complete combustion means that the limiting reagent is carbon, so most of the coke is removed and more energy is generated. Therefore, complete combustion is more efficient than partial combustion but deciding which type will be used is highly dependent on the FCC feed quality and the equipment owned by the refinery [15]. Assuming complete combustion is chosen as the removal method in the regenerator, the coke composition on the catalyst surface is reduced down to 0.10%.

From the regenerator, the catalyst flows down through the standpipe which provides the proper pressure drop for the catalyst to be transported down the FCC unit. These standpipes are designed for the desired flux rate of 250 lb/s/ft^2 making sure the catalyst gets through the plug valve that leads to the riser [14].

Similarly, flue gas exits through the top of the regenerator liberating a substantial amount of energy. To recover this energy, oil refineries have opted for sending the flue gas to a reboiler where most of the heat is used to produce high pressure steam. Other techniques require the

implementation of shell & tube heat exchangers that will exchange the flue gas with boiling water, producing steam as a result [18]. When this process takes place, the flue gas experiences a pressure drop that occurs when the gas passes through the system valves. To achieve pressure recovery, a turbo expander must be used which subsequently recovers power. The turbo expander is designed with four different parts: the expander, the generator, the air blower, and the steam turbine [16]. The flue gas power recovery scheme can be seen in figure 14 below.

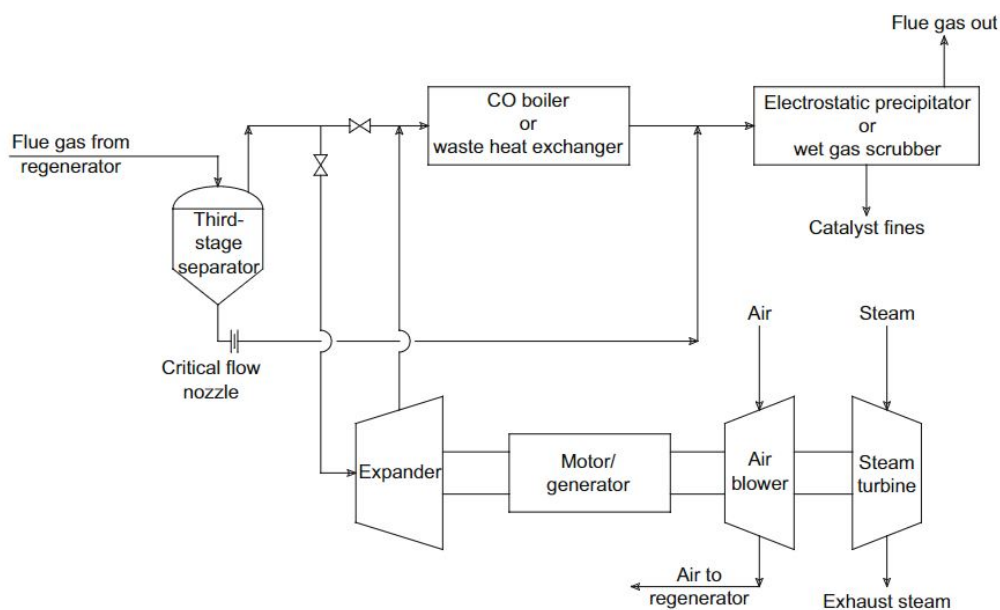


Figure 14 - Power Recovery Scheme [15]

In the expander section a steam turbine is used to generate power (electricity) when mixing with the flue gas. The motor or generator is used to remove or provide power that may be needed for efficient transportation of the gas through the system. It is in the generator where the thermal energy is recovered [13].

The vapors that were separated from the catalyst in the reactor are sent to the main fractionator. In this unit, the liquid products are recovered from the vapors through condensation and re-vaporization of the hydrocarbon components. The main fractionator operates similarly to

the crude tower since distillation takes place within it [15]. Nevertheless, before the feed is treated it is necessary to cool down the vapors before fractionating it. Additionally, it is imperative to keep in mind that further separation will be required because a large quantity of gases will go overhead with the gasoline [13].

To cool down the vapors, these go through a series of pumps that wash out any remaining catalyst and take them to the pool quench. The desired temperature in the pool quench is about 360°C because at this temperature the formation of coke is reduced [17].

After the vapors are fractionated, the bottoms of the column are recovered and is known as slurry oil. This slurry oil is low in sulfur, metals, and ash; thus it can be used or sold as a carbon feedstock [15]. The rest of the fractionated vapors is separated into heavy cycle oil (HCO), light cycle oil (LCO), and heavy naphtha. The HCO is typically recycled back to the reactor for further cracking and is rarely recovered as a product. The heavy naphtha may be blended with gasoline or otherwise used as an absorption gas oil in the plant. Finally, the LCO is our desired product. The LCO is treated by a stripper to produce gasoline and diesel; while it is also blended into the heating oil pool after being treated by the sulfur removal to reduce the amount of sulfur present in the oil [18].

The unstabilized gasoline and all the other lighter hydrocarbons leave as vapor through the top of the main fractionator. These vapors are condensed forming a mixture of liquid and vapor hydrocarbons with water. This mixture is finally separated in the overhead drum. When this separation takes place, the hydrocarbons are split into liquid and vapor hydrocarbons (in different streams) [16]. The vapor hydrocarbons are sent to a wet gas compressor (WGC), where the heaviest hydrocarbons of the mixture (C3 and C4) are condensed, while the lighter hydrocarbons remain in the vapor phase.

Similarly, the liquid hydrocarbons are pumped back and serve as the reflux for the distillation column. Since water is in its liquid state, it is also refluxed back along with the liquid hydrocarbons [14].

The final section of the FCC unit is known as the gas plant. The gas plant starts with the wet gas compressor. This unit consists of a two stage compressor whose driving force is high pressured steam [17]. When the overhead vapors enter the compressor, they are partially condensed and then sent to the interstage drum. Once they pass through the drum, the vapors reach the second stage of the compressor. It is here where they are discharged to the high pressure separator (HPS) through a cooler [15]. The vapors that are in the HPS are sent to the primary absorber, while the liquid in the HPS are sent to the stripper. In the primary absorber, the C3's and heavier compounds are recovered and then absorbed in the oil while the vapors are sent to the secondary absorber. In the secondary absorber, the gasoline that was not absorbed by the primary absorber is recovered [17].

On the other hand, the liquids of the HPS are sent to the stripper or de-ethanizer. In the stripper, the C2's, hydrogen sulfide (H₂S), and entrained water are separated from the C3 and heavier hydrocarbons, producing naphtha which goes to the debutanizer [16].

In the debutanizer, the C3 and C4's compounds are separated from the gasoline, which is further treated in the water wash system (WWS) for removal of sulfur and other organic nitrogen compounds by reacting these with ammonia to produce urea [14].

- The Chosen Sulfur treatment:

There are a variety of techniques used in order to reduce sulfur in gasoline which include pre-treatment, in suit treatment and post-treatment [36]. Reactive adsorption desulfurization (RADS) is the chosen technique to achieve the goal of this process. Sulfur content in FCC gasoline can be reduced to less than or equal to 10 ppm using RADS process while minimizing octane loss. The process consists of deflurization and adspotion process [37]. Furthermore, RADS process describes the reactive adsorption and capture of sulfur using metal-base solid sorbent at high temperatures and in the presence of hydrogen gas at low pressure[38]. One of the required factors for RADS process in order to ensure effective performance is the catalyst's specifications. Sorbent must have high desulfurization activity; thus, Ni/ZnO is considered an ideal option[37].

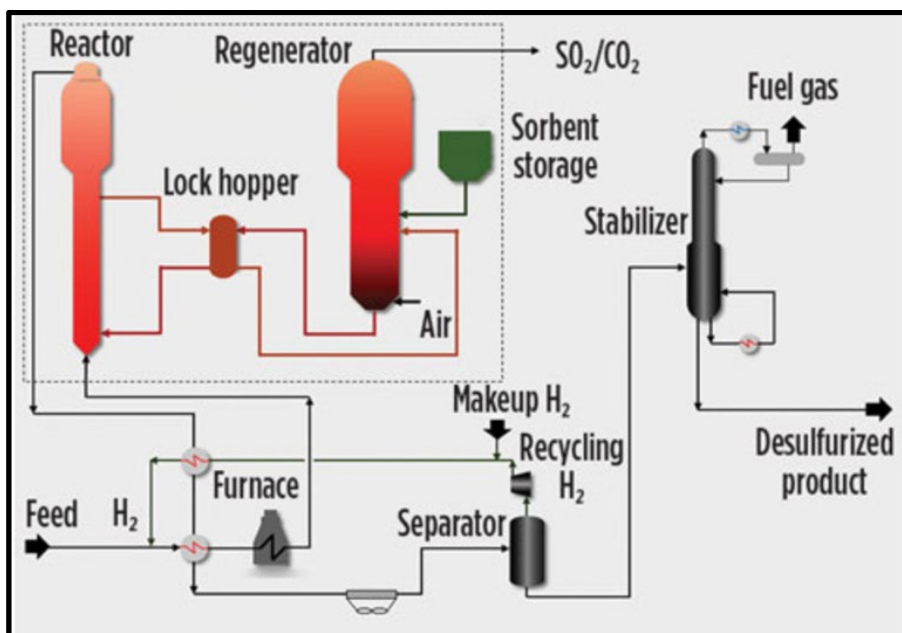


Figure 15 : Simplified PFD of RADS process [38]

After the FCC gasoline is produced, the RADS process takes places as shown in figure 15. The first step is to mix hydrogen gas with the FCC gasoline which contains approximately

300 ppm-1000 ppm sulfur content[38]. Heating the mixture via the furnace before going to the reactor is required in order for it to correspond with the reactor temperature which is approximately **420 °C- 440 °C** [38]. Regarding hydrogen gas, it can be obtained from either steam reforming process or as recovery from refinery off gases. Inside the reactor, the adsorption desulfurization reaction occurs which is shown in figure 16.

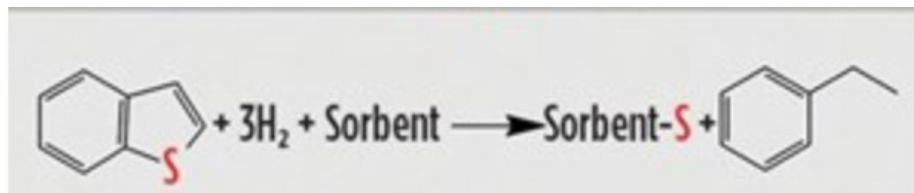


Figure 16: The reaction of Reactive adsorption desulfurization (RADS) [39]

The presence of hydrogen in the adsorption desulfurization process plays a significant role as it helps to extract the sulfur being captured by the sorbent. In addition, it helps reduce coking. The reaction occurs in the gas phase in order to prevent the porosity from having the remaining of the oil [38].

After the reactor, the used sorbent is transferred for sorbent regeneration process where oxidation reaction will occur before recycling the sorbent. the clean stream, is transferred from the top of the reactor and then cooled before entering the separator where hydrogen will be separated from the gasoline. Hydrogen gas leaves from the top of the separator and is recycled back to the feed while the gasoline enters the stabilizer column. The stabilizer column will operate as a stripper where fuel gas leaves from the top and is recycled back to the stabilizer. In addition, the product which is the desulfurized gasoline leaves from the bottom of the stabilizer column. Besides the fact that sulfur content in the produced gasoline is below 10 ppm and the loss of octane is minimized to less than 0.5, the yield of the product can be approximated to

99.5%.[38]. As a result, the RADS process is considered the ideal technology for the purpose of this project.

Appendix I (Design Basis)

Based on the market analysis, we chose to place our plant in Texas, United States. This is because Texas is very close to our crude oil supplier located in the Gulf of Mexico which cuts down the cost of the transportation and increases profitability [44].

The main goal of this design is to produce approximately 12 thousand barrels of gasoline per day with a sulfur concentration lower than 10 ppm to meet environmental regulations. The operating conditions of the FCC unit is at a temperature of 930-1050 °F with an outlet vapor velocity in the riser between 40-60 ft/s. The catalyst we are using is zeolite and the reactor will operate at a pressure of 1.7 bar [14].

Appendix II (BFD and Material Balance)

- BFD and overall mass balance of the FCC unit:

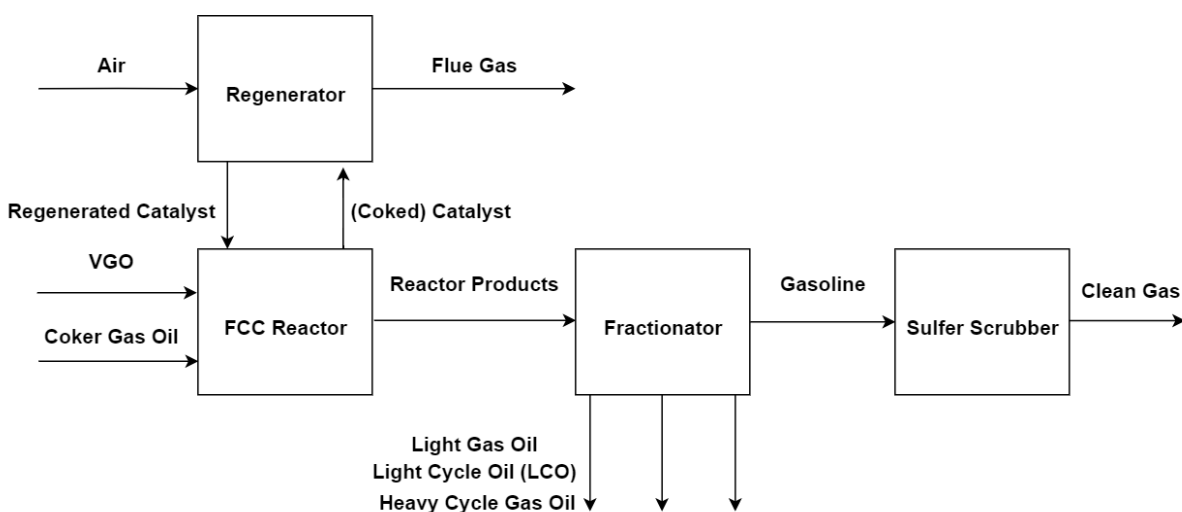


Figure 17: BFD of the FCC unit [1]

Table 1: Rough Overall Mass Balance in FCC Unit

Component	Mass In (lb/hr)	Mass Out (lb/hr)
VGO	790,486	0
Coker Off Gas	10,775	0
Tail Gas	0	57,465
LPG	0	112,357
Gasoline	0	391,166
LCO	0	161,920
Slurry	0	55,505
Overall Mass Balance	801,260	778,413

In refineries, the rough material material balance is considered accurate if the mass in and out differ within a range of 97 to 103 %. Given that our weight balance falls within the 97% range, this weight balance estimate is considered above industry average [16].

- **BFD of the Desulfurization unit (RADS) and overall mass balance:**

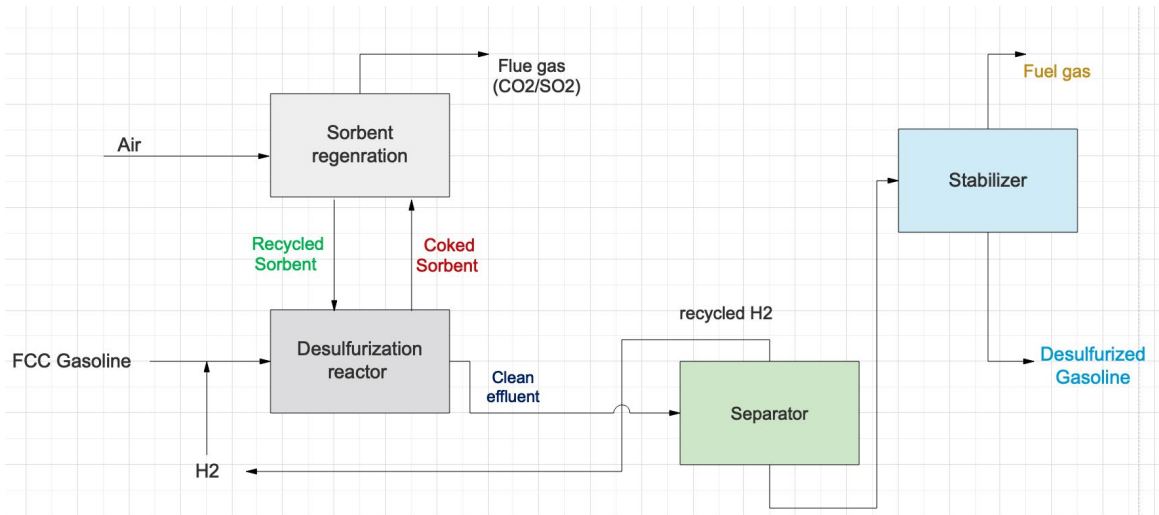


Figure 18: BFD of the desulfurization unit (RADS)

Appendix III: PFD

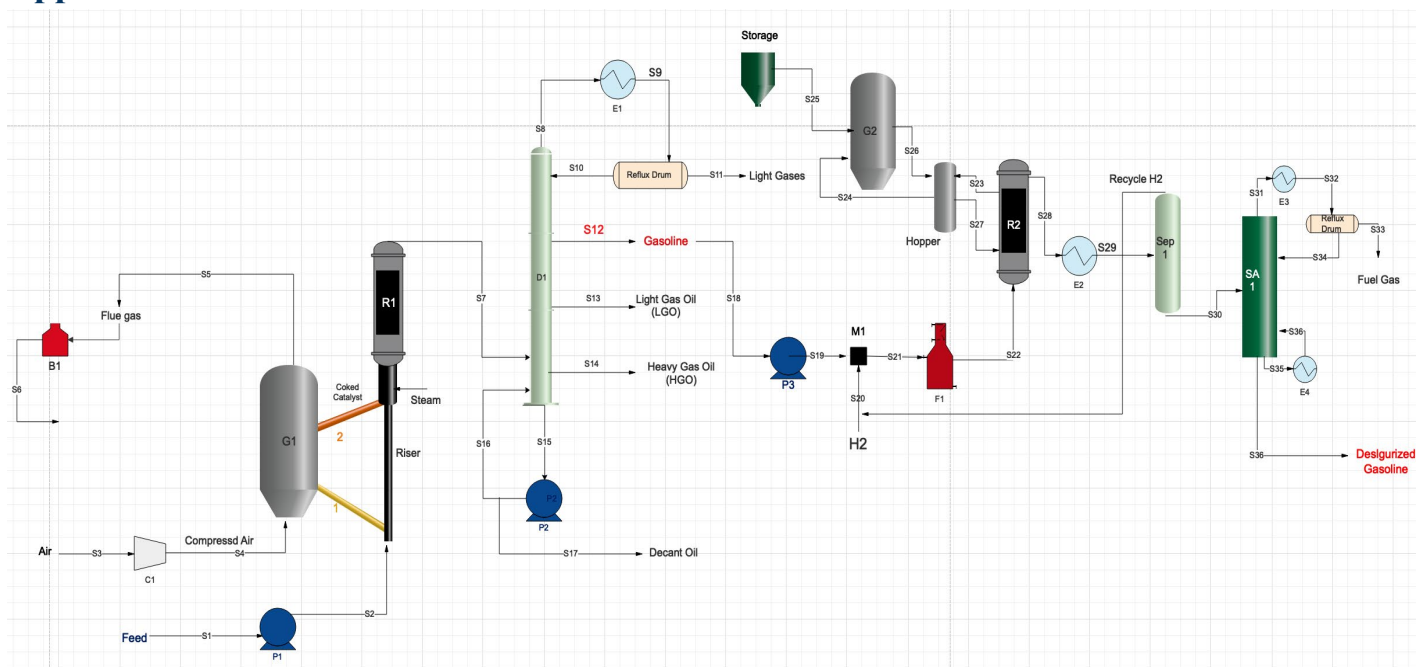


Figure 19 : Simple process flow diagram of FCCU and RADSU [12][38]

Explanation of all equipment in the PFD shown in figure 19:

1. **Pump:** A device used for either raising or transference of liquids using pressure or suction
 - **Pump 1 (P1):** Used to transfer the feedstock into the reactor
 - **Pump 2 (P2):** Used to transfer the Decant oil as recycle back to the fractionator
 - **Pump 3 (P3):** Used to transfer the produced FCC gasoline to the RADS unit.
2. **Reactor:** A process vessel where chemical reactions take place
 - **Reactor 1 (R1):** a riser reaction where the cracking reaction is carried out in the riser before being transferred upward to the reactor where coked catalyst is separated from the products.
 - **Reactor 2 (R2):** desulfurization reactor where the adsorption desulfurization reaction is carried out.
3. **Regenerator:** A system used to recycle/return a component to its strength
 - **Regenerator 1 (G1):** fluidized bed regenerator used to burn coke from the coked catalyst coming from the reactor in order to reactivate the catalyst
 - **Regenerator 2 (G2):** Sorbent regenerator where the spent sorbent is sent.
4. **Compressor:** A device used to increase the pressure of gas

- **Compressor 1 (C1)**: Air compressor used to increase low pressure air to higher than atmospheric pressure.
- 5. **Boiler**: An apparatus used to convert liquid to a vapor under high temperature.
 - **Boiler 1 (B1)**: CO boiler used to recover the energy(heat) from fuel gas and convert CO to CO₂ in the fuel gas.
- 6. **Column**: Equipment used to separate a mixture into multiple components or fractions
 - **Fractionator 1(D1)**: Separator used to separate the effluent coming from the reactor into multiple components based on their boiling point.
 - **Separator 1 (Sep1)**: Separator used to separate the hydrogen gas from the effluent coming from the reactor.
 - **Stabilizer 1 (SA1)**: used to separate the fuel gas from the desulfurized gasoline
- 7. **Heat Exchanger**:
 - **Heat Exchanger 1 (E1)**: a system used to condense the light gases coming from the fractionator
 - **Heat Exchanger 2 (E2)**: a system used to cool the effluent coming from the reactor going to the separator
 - **Heat Exchanger 3 (E3)**: a system used to condense other fuel gases coming from the stabilizer before going the reflux drum
 - **Heat Exchanger 4 (E4)** a system used to heat the bottom effluent before going back as recycled to the stabilizer
- 8. **Furnace**: Equipment used to higher temperature of flow in order to provide heat for the needed process.
 - **Furnace 1 (F1)**: Used to heat up the mixture of gasoline and hydrogen before going to the reactor
- 9. **Tank**: Containers used to storage chemicals
 - **Tank 1 (Storage)**: Storage container for sorbent provided regenerator
- 10. **Reflux Drum**: A vessel used to provide cooling and condensation for the condensed reflux coming from the top of column before returning back to the column
 - **Drum 1 (reflux drum 1)**: used to provide surge between the condenser and fractionator
 - **Drum 2 (reflux drum 2)**: used to provide surge between the condenser and stabilizer
- 11. **Lock Hopper**: a system used to control the recycled sorbents

- **Lock Hopper 1 (Hopper)**: used between the reactor and regenerator to control the recycled sorbent and its needed flow rate for the adsorption reaction
- 12. Mixer:** a system used in order to mix different fluids regardless the phases
- **Mixer 1 (M1)**: used to mix the liquid gasoline coming from the FCC unit with hydrogen gas

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